

UNIT – 3

ASSESSMENT – 3

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Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet.

1. Propositional Logic

Let:

- = It is raining.
- = The ground is wet.

Statement 1:

Statement 2:

Goal:

2. Convert to CNF

The implication:

becomes:

Therefore, our clauses are:

Clause 1:

Clause 2:

3. Negate the Goal

Goal:

Negated goal:

Add it to the clauses.

Clause 3:

4. Resolution

Resolve:

with:

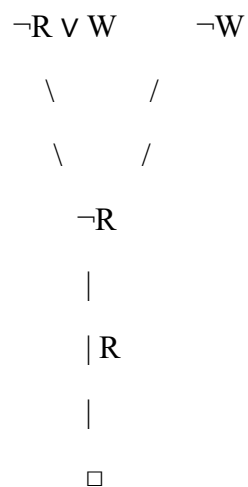
Resolving on :

Now we have:

Resolve this with:

Therefore:

Resolution Tree



Conclusion

Since the **Empty Clause** ($\square$) is derived, the negation of the goal leads to a contradiction.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks.

1. Propositional Logic

Let:

- = Rahul submitted the assignment.
- = Rahul receives internal marks.

Statement 1:

Statement 2:

Goal:

2. CNF

Convert:

to:

Therefore:

Clause 1:

Clause 2:

3. Negate Goal

Goal:

Negated goal:

Clause 3:

4. Resolution Steps

Resolve:

with:

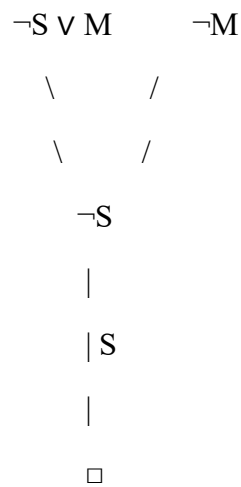
Result:

Now resolve:

with:

Result:

Resolution Tree



Conclusion

The Empty Clause is obtained.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books.

1. Propositional Logic

Let:

- = Priya is a library member.
- = Priya can borrow books.

Statement 1:

Statement 2:

Goal:

2. CNF

Convert the implication:

to:

Therefore:

Clause 1:

Clause 2:

3. Negate Goal

Clause 3:

4. Resolution

Resolve:

with:

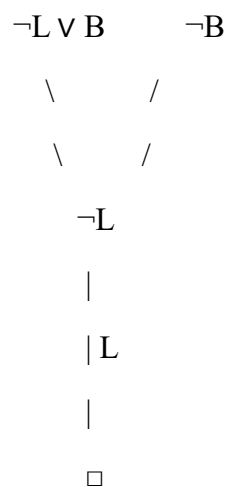
Result:

Now resolve:

with:

Result:

Resolution Tree



Final Conclusion

Since the Empty Clause is derived:

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement.

1. Propositional Logic

Let:

- = Arun cleared the aptitude test.
- = Arun is eligible for placement.

Statement 1:

Statement 2:

Goal:

2. CNF Clauses

Convert:

to:

Therefore:

Clause 1:

Clause 2:

3. Negate the Goal

Clause 3:

4. Resolution Steps

Resolve:

and:

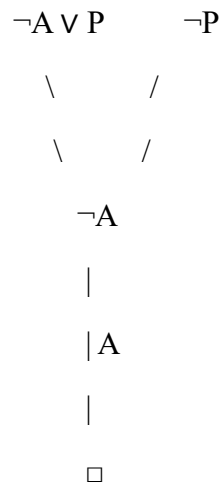
Result:

Now resolve:

and:

Result:

Resolution Tree



Final Conclusion

The Empty Clause is derived.

Question 5 – Access Control System

This one has **two implications**, so it requires an additional resolution step.

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access.

1. Propositional Logic

Let:

- = User entered the correct password.

- = User is authenticated.
- = User is granted access.

Statements:

Goal:

2. Convert to CNF

Statement 1

becomes:

Clause 1:

Statement 2

becomes:

Clause 2:

Statement 3

Clause 3:

3. Negate the Goal

Goal:

Negated goal:

Clause 4:

4. Resolution Steps

Step 1

Resolve:

with:

Resolving on :

Step 2

Resolve:

with:

Resolving on :

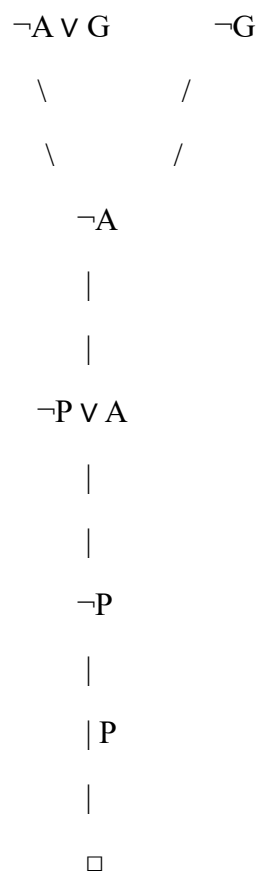
Step 3

Resolve:

with:

Result:

Resolution Tree



Final Conclusion

Since the Empty Clause $\square$ is derived:

The logical chain is:

Correct Password

↓

Authenticated

↓

Granted Access

Quick Summary for Your Assessment

Question	Goal	Empty Clause?	Result
1. Rain	Ground is wet	☐	Proved
2. Assignment	Rahul receives marks	☐	Proved
3. Library	Priya can borrow books	☐	Proved
4. Placement	Arun is placement eligible	☐	Proved
5. Access Control	User gets access	☐	Proved

SOURCE CODE

1. # Resolution Algorithm

Question 1: Rain and Wet Ground

```
print("=====")
```

```
print(" RESOLUTION ALGORITHM")
```

```
print(" Rain and Wet Ground")
```

```
print("=====")
```

```
# Knowledge Base:
```

```
# 1. If it rains, ground becomes wet.
```

R -> W

CNF: $\sim R$ OR W

#

2. It is raining.

R

#

Goal:

W

print("\nKnowledge Base:")

print("1. R -> W")

print("2. R")

print("\nCNF Clauses:")

print("C1 = $\sim R$ OR W")

print("C2 = R")

print("\nGoal: W")

Negate the goal

print("Negated Goal: $\sim W$ ")

print("\n--- Resolution Steps ---")

Step 1

```
print("\nStep 1:")  
print("C1 =  $\sim R$  OR W")  
print("C3 =  $\sim W$ ")  
print("Resolving on W...")  
print("Result =  $\sim R$ ")
```

Step 2

```
print("\nStep 2:")  
print("C2 = R")  
print("C4 =  $\sim R$ ")  
print("Resolving on R...")  
print("Result = EMPTY CLAUSE ( $\square$ )")
```

```
print("\n=====")  
print("CONCLUSION")  
print("=====")  
print("Empty Clause ( $\square$ ) is derived.")  
print("Therefore, the ground is wet is PROVED.")
```

2. # Resolution Algorithm

Question 2: Student Assignment Submission

```
print("=====")
print(" RESOLUTION ALGORITHM")
print(" Student Assignment Submission")
print("=====")
```

Knowledge Base:

$S \rightarrow M$

S

#

Goal:

M

```
print("\nKnowledge Base:")
```

```
print("1.  $S \rightarrow M$ ")
```

```
print("2. S")
```

```
print("\nCNF Clauses:")
```

```
print("C1 =  $\sim S \text{ OR } M$ ")
```

```
print("C2 = S")
```

```
print("\nGoal: M")
```

Negated goal

```
print("Negated Goal:  $\sim M$ ")
```

```
print("\n--- Resolution Steps ---")
```

Step 1

```
print("\nStep 1:")
```

```
print("C1 =  $\sim$ S OR M")
```

```
print("C3 =  $\sim$ M")
```

```
print("Resolving on M...")
```

```
print("Result =  $\sim$ S")
```

Step 2

```
print("\nStep 2:")
```

```
print("C2 = S")
```

```
print("C4 =  $\sim$ S")
```

```
print("Resolving on S...")
```

```
print("Result = EMPTY CLAUSE ( $\square$ )")
```

```
print("\n=====")
```

```
print("CONCLUSION")
```

```
print("=====")
```

```
print("Empty Clause ( $\square$ ) is derived.")
```

```
print("Therefore, Rahul receives internal marks is PROVED.")
```

3. # Resolution Algorithm

Question 3: Library Membership

```
print("=====")
```

```
print("  RESOLUTION ALGORITHM")
```

```
print("  Library Membership")
```

```
print("=====")
```

```
# Knowledge Base:
```

```
# L -> B
```

```
# L
```

```
#
```

```
# Goal:
```

```
# B
```

```
print("\nKnowledge Base:")
```

```
print("1. L -> B")
```

```
print("2. L")
```

```
print("\nCNF Clauses:")
```

```
print("C1 = ~L OR B")
```

```
print("C2 = L")
```

```
print("\nGoal: B")
```

```
# Negated goal
```

```
print("Negated Goal: ~B")
```

```
print("\n--- Resolution Steps ---")
```

Step 1

print("\nStep 1:")

print("C1 = $\sim$ L OR B")

print("C3 = $\sim$ B")

print("Resolving on B...")

print("Result = $\sim$ L")

Step 2

print("\nStep 2:")

print("C2 = L")

print("C4 = $\sim$ L")

print("Resolving on L...")

print("Result = EMPTY CLAUSE ($\square$)")

print("\n=====")

print("CONCLUSION")

print("=====")

print("Empty Clause ($\square$) is derived.")

print("Therefore, Priya can borrow books is PROVED.")